3 Propan-2-ol, $(CH_3)_2CHOH$, is sometimes added to fuel to help it burn.

Fig. 3.1 shows some reactions of propan-2-ol.

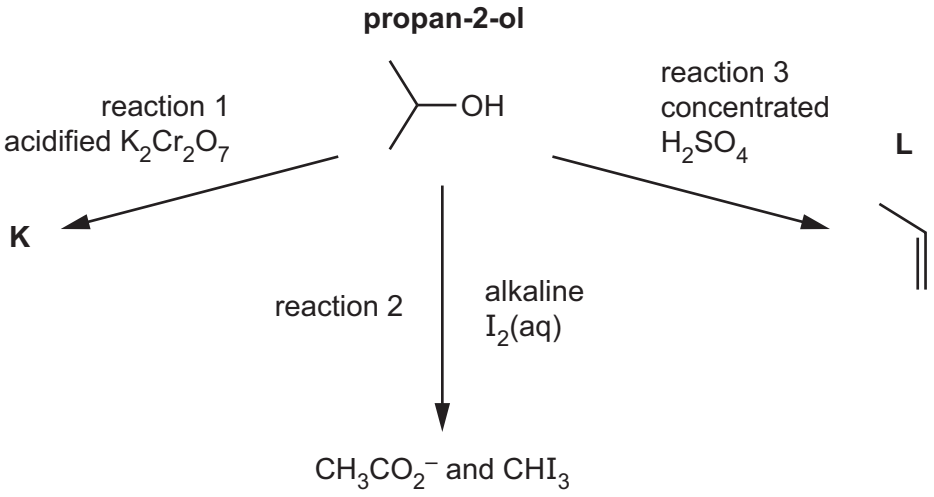


Fig. 3.1

(a) (i) Draw the structure of organic compound **K**.

[1]

(b) Propan-2-ol reacts with sodium to produce $(CH_3)_2CH–O^-$ anions.

These anions react with 2-bromopropane to form compound **N**, as shown in Fig. 3.3.

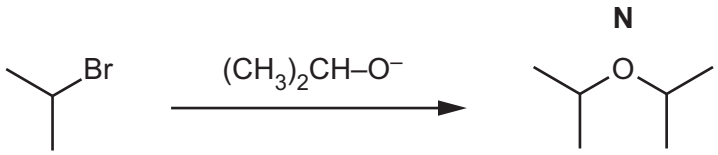


Fig. 3.3

(i) Write an equation for the reaction of propan-2-ol with sodium.

..... [1]

(ii) The reaction of 2-bromopropane with $(CH_3)_2CH–O^-$ anions follows an S_N1 mechanism.

Complete Fig. 3.4 to show this mechanism. Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.



Fig. 3.4

[3]

(iii) Suggest how the rate of the S_N1 reaction would change, if at all, if 2-chloropropane were used instead of 2-bromopropane.

Explain your answer.

.....

.....

..... [2]

4 Compounds **P** and **Q** are structural isomers.



Fig. 4.1

(a) **(i)** Define structural isomerism.

.....

..... [2]

(b) Fig. 4.3 shows a synthesis starting from **T**, a different isomer of **R** and **S**.

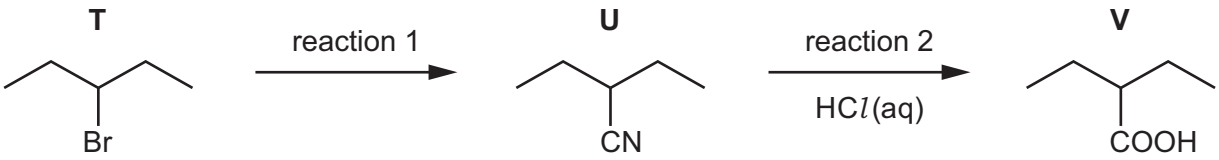


Fig. 4.3

(i) Identify the reagent and conditions for reaction 1.

..... [1]

3 Cycloalkanes show similar chemical properties to alkanes but have the same empirical formula as alkenes.

(a) Define empirical formula.

.....

..... [1]

(d) Fig. 3.2 shows a reaction cycle involving cyclopentane, cyclopentene and C_5H_9Cl.

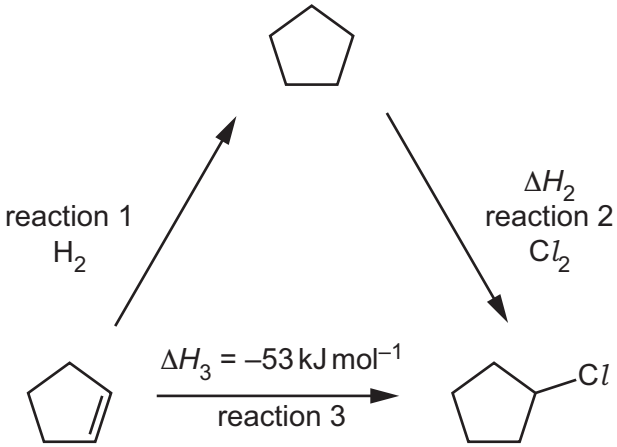


Fig. 3.2

(i) Identify a suitable reagent for reaction 3.

..... [1]

Table 3.3

bond	functional groups containing the bond	characteristic infrared absorption range (in wavenumbers) / cm^{-1}
C–O	hydroxy, ester	1040–1300
C=C	aromatic compound, alkene	1500–1680
C=O	amide carbonyl, carboxyl ester	1640–1690 1670–1740 1710–1750
C≡N	nitrile	2200–2250
C–H	alkane	2850–2950
N–H	amine, amide	3300–3500
O–H	carboxyl hydroxy	2500–3000 3200–3650

[Total: 13]

4 Fig. 4.1 shows a possible synthesis of propene, C_3H_6.

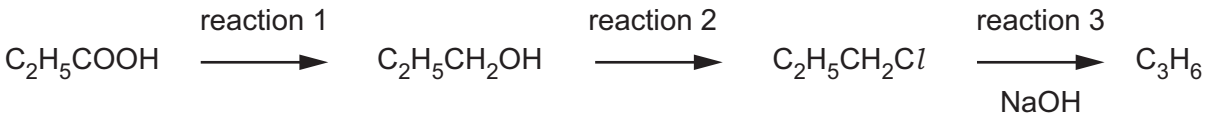


Fig. 4.1

(a) (i) Identify the type of reaction that occurs in reaction 1.

..... [1]

(ii) Suggest a suitable reagent for reaction 2.

..... [1]

(iii) Reaction 3 is an elimination reaction.

Write an equation for this reaction and identify the solvent and conditions used.

equation .....

solvent and conditions .....

[2]

4 Fig. 4.1 shows how propane, C_3H_8, can be converted to propanoic acid, CH_3CH_2COOH.

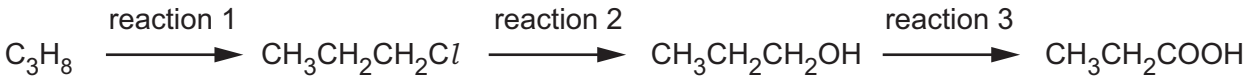


Fig. 4.1

(a) Reaction 1 in Fig. 4.1 takes place in the presence of sunlight.

$$C_3H_8 + Cl_2 \rightarrow CH_3CH_2CH_2Cl + HCl$$

The reaction takes place via initiation, propagation and termination steps.

(i) Name the mechanism shown by reaction 1.

..... [1]

(b) State the reagent and solvent required for reaction 2.

..... [1]

3 Cycloalkanes show similar chemical properties to alkanes but have the same empirical formula as alkenes.

(a) Define empirical formula.

.....

..... [1]

(d) Fig. 3.2 shows a reaction cycle involving cyclopentane, cyclopentene and C_5H_9Cl.

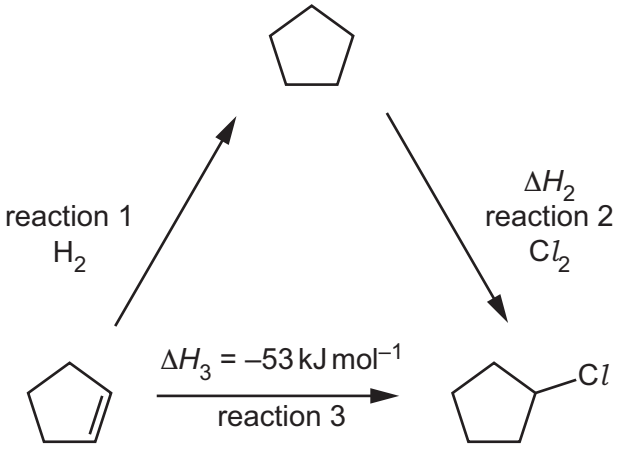


Fig. 3.2

(i) Identify a suitable reagent for reaction 3.

..... [1]

Table 3.3

bond	functional groups containing the bond	characteristic infrared absorption range (in wavenumbers) / cm^{-1}
C–O	hydroxy, ester	1040–1300
C=C	aromatic compound, alkene	1500–1680
C=O	amide carbonyl, carboxyl ester	1640–1690 1670–1740 1710–1750
C≡N	nitrile	2200–2250
C–H	alkane	2850–2950
N–H	amine, amide	3300–3500
O–H	carboxyl hydroxy	2500–3000 3200–3650

[Total: 13]

4 Fig. 4.1 shows a possible synthesis of propene, C_3H_6.

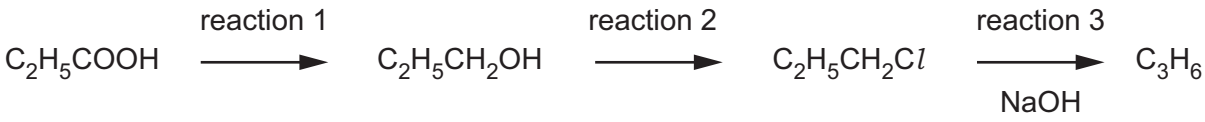


Fig. 4.1

(a) (i) Identify the type of reaction that occurs in reaction 1.

..... [1]

(ii) Suggest a suitable reagent for reaction 2.

..... [1]

(iii) Reaction 3 is an elimination reaction.

Write an equation for this reaction and identify the solvent and conditions used.

equation .....

solvent and conditions .....

[2]

4 The skeletal formula of **E** is shown in Fig. 4.1.

E

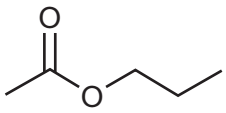


Fig. 4.1

(a) State the molecular formula of **E**.

..... [1]

(iii) CH_3OH is used to make CH_3CN in a two-step process, as shown in Fig. 4.2.

$$CH_3OH \xrightarrow{\text{step 1}} CH_3Cl \xrightarrow{\text{step 2}} CH_3CN$$

Fig. 4.2

State the reagents and conditions required for steps 1 and 2.

step 1 .....

step 2 .....

[3]

6 **(a)** $HOCl$ reacts with $CH_2{=}CH_2$ to produce $HOCH_2CH_2Cl$ in an electrophilic addition reaction.

(i) Define addition reaction.

.....

..... [1]

(c) $HOCH_2CH_2Cl$ reacts in a two-step synthesis to produce $HOCH_2CH_2COOH$, as shown in Fig. 6.2.

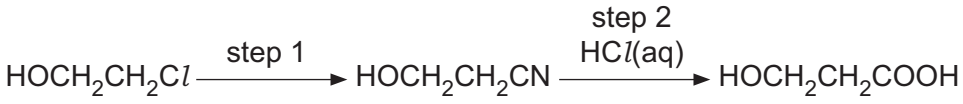


Fig. 6.2

(i) State the reagent and conditions for step 1.

..... [1]

4 **(a)** Propanoic acid, CH_3CH_2COOH, reacts with reducing agent **Q** to produce propan-1-ol, $CH_3CH_2CH_2OH$.

(i) Suggest the formula of reducing agent **Q**.

..... [1]

(b) Propan-1-ol is converted to compound **T** in a three-step synthesis.

In step 1, $SOCl_2$ is added to propan-1-ol to produce compound **E**.

In step 2, **E** reacts with a suitable reagent to produce butanenitrile.

In step 3, butanenitrile is heated with NaOH(aq).

(i) Name the type of reaction that occurs in step 1.

..... [1]

(ii) Identify the reagent and conditions required in step 2.

..... [1]

[Total: 9]

Table 4.1

bond	functional groups containing the bond	characteristic infrared absorption range (in wavenumbers) / cm^{-1}
C−O	hydroxy, ester	1040–1300
C=C	aromatic compound, alkene	1500–1680
C=O	amide carbonyl, carboxyl ester	1640–1690 1670–1740 1710–1750
C≡N	nitrile	2200–2250
C−H	alkane	2850–2950
N−H	amine, amide	3300–3500
O−H	carboxyl hydroxy	2500–3000 3200–3650

5 Compound **A** contains the elements carbon, hydrogen and oxygen only.
When **A** is heated with $H_2SO_4(aq)$, compounds **B** and **C** are produced, as shown in Fig. 5.1.

$$\mathbf{A} + 2H_2O \xrightarrow[\text{heat}]{H_2SO_4(aq)} \underset{\mathbf{B}}{HOOCCH_2COOH} + \underset{\mathbf{C}}{2CH_3CH_2OH}$$

Fig. 5.1

(a) Draw the structure of **A**.

[2]

(d) A student suggests a two-step synthesis to produce **B**, as shown in Fig. 5.2.

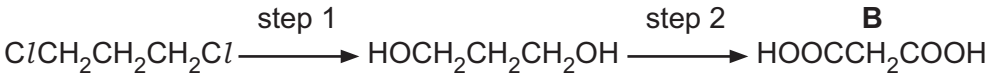


Fig. 5.2

(i) Identify the reagents and conditions required in steps 1 and 2.

step 1 .....

step 2 .....

[2]

1 Phosphorus and chlorine are elements in Period 3 of the Periodic Table.

(a) Chlorine forms three different compounds with phosphorus.

The most common compounds are PCl_3 and PCl_5.

(i) Complete Table 1.1.

Table 1.1

compound	oxidation number of P	oxidation number of Cl
PCl_3		
PCl_5		

[2]

(b) Fig. 1.1 shows a reaction scheme involving P_4 and Cl_2.

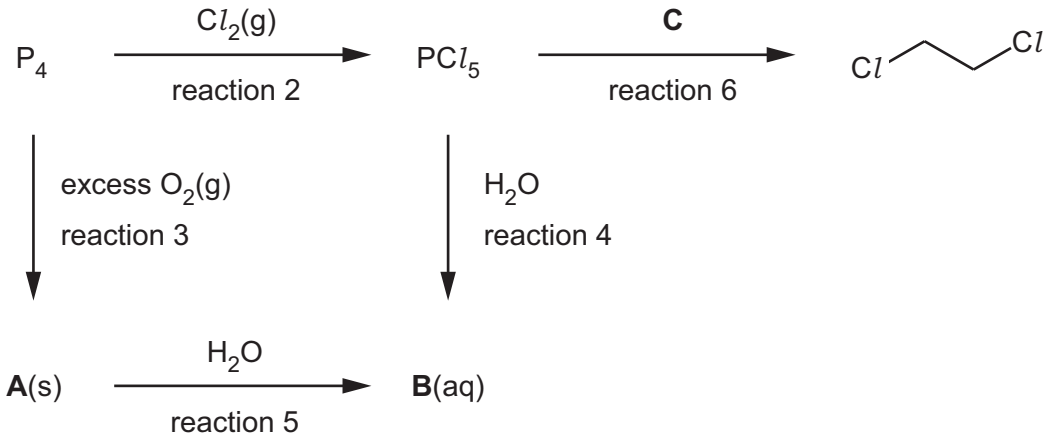


Fig. 1.1

(i) Suggest what is observed in reaction 2.

..... [1]

(iv) Name **B**, which is formed in both reactions 4 and 5.

..... [1]

(v) Draw the structure of **C**, used in reaction 6.

[1]

(c) 1,2-dichloroethane, $ClCH_2CH_2Cl$, reacts with NaOH to produce an unsaturated compound, C_2H_3Cl, as shown in reaction 7.

reaction 7 $\quad ClCH_2CH_2Cl + NaOH \rightarrow C_2H_3Cl + H_2O + NaCl$

(i) State what is meant by unsaturated.

.....

..... [1]

(ii) State the conditions for reaction 7.

..... [1]

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2 Chlorine is one of the elements in Group 17 of the Periodic Table.

(a) **(i)** Describe the colours of the Group 17 elements, chlorine to iodine, at room temperature.

.....

..... [1]

(ii) State the essential condition for chlorine to react with C_2H_6 at room temperature.

..... [1]

(iv) Aqueous $AgNO_3$ dissolved in ethanol reacts with an aqueous solution of $CHCl_3$.

State what is observed in this reaction. Explain your answer.

.....

.....

..... [2]

4 Bromoalkanes are used widely in industry, although there is increasing concern about their environmental impact.

Fig. 4.1 shows a reaction scheme involving 1,2-dibromoethane.

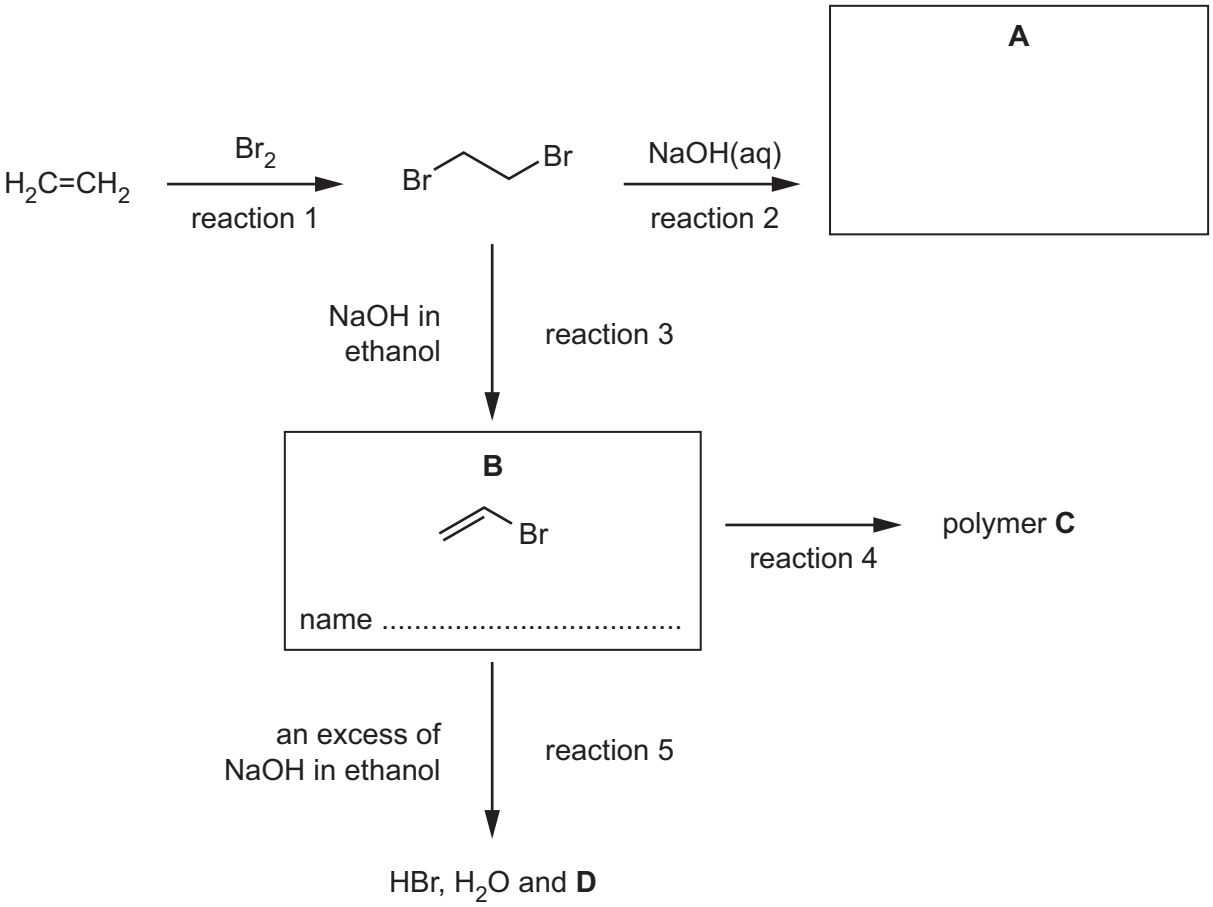


Fig. 4.1

(a) Complete Fig. 4.2 to show the mechanism for the formation of 1,2-dibromoethane in reaction 1.

Include charges, dipoles, lone pairs of electrons and curly arrows as appropriate.



Fig. 4.2

[3]

(c) **(i)** Complete Fig. 4.1 to:

- draw the structure of compound **A**
- name compound **B**.

[2]

(iii) In reaction 5, compound **B** reacts with an excess of NaOH dissolved in ethanol. The products are HBr, H_2O and an unsaturated hydrocarbon **D**.

Suggest the identity of **D**.

..... [1]

BLANK PAGE

2 Chlorine is one of the elements in Group 17 of the Periodic Table.

(a) (i) Describe the colours of the Group 17 elements, chlorine to iodine, at room temperature.

.....

..... [1]

(ii) State the essential condition for chlorine to react with C_2H_6 at room temperature.

..... [1]

(iv) Aqueous $AgNO_3$ dissolved in ethanol reacts with an aqueous solution of $CHCl_3$.

State what is observed in this reaction. Explain your answer.

.....

.....

..... [2]

4 **(a)** $NH_3(g)$ reacts with $HCl(g)$ to produce $NH_4Cl(s)$, as shown.

$$NH_3(g) + HCl(g) \rightarrow NH_4Cl(s)$$

Draw a diagram to show the ionic, covalent and coordinate bonding present in a formula unit of NH_4Cl.

[2]

(c) Substitution reactions of NH_3 and OH^- with halogenoalkanes both involve a lone pair of electrons.

(i) Name the role of NH_3 and OH^- in these reactions.

..... [1]

(ii) Suggest which species, NH_3 or OH^-, is more reactive during these reactions. Explain your answer.

.....

.....

..... [1]

(d) When 2-bromo-2-methylpropane reacts with OH^-, two mechanisms, S_N1 and S_N2, both occur. The S_N2 mechanism has a slower rate.

Fig. 4.1 shows the reaction pathway diagram for the S_N1 mechanism.

Sketch a graph on Fig. 4.1 to show the reaction pathway for the S_N2 mechanism.

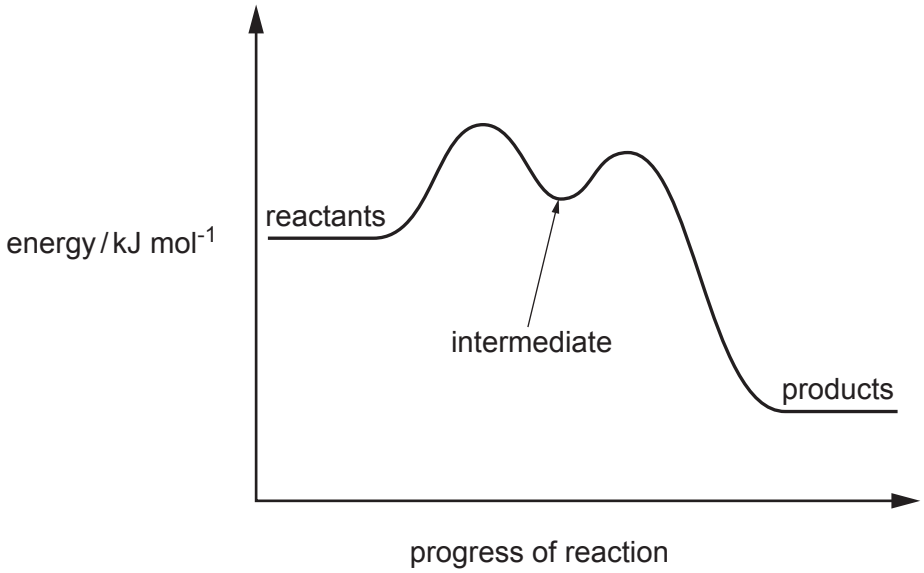


Fig. 4.1

[2]

(e) **(i)** Complete Fig. 4.2 to show the mechanism for the S_N1 reaction that occurs when $CH_3CHBrC_2H_5$ reacts with NH_3 to produce $CH_3CH(NH_2)C_2H_5$.
Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.

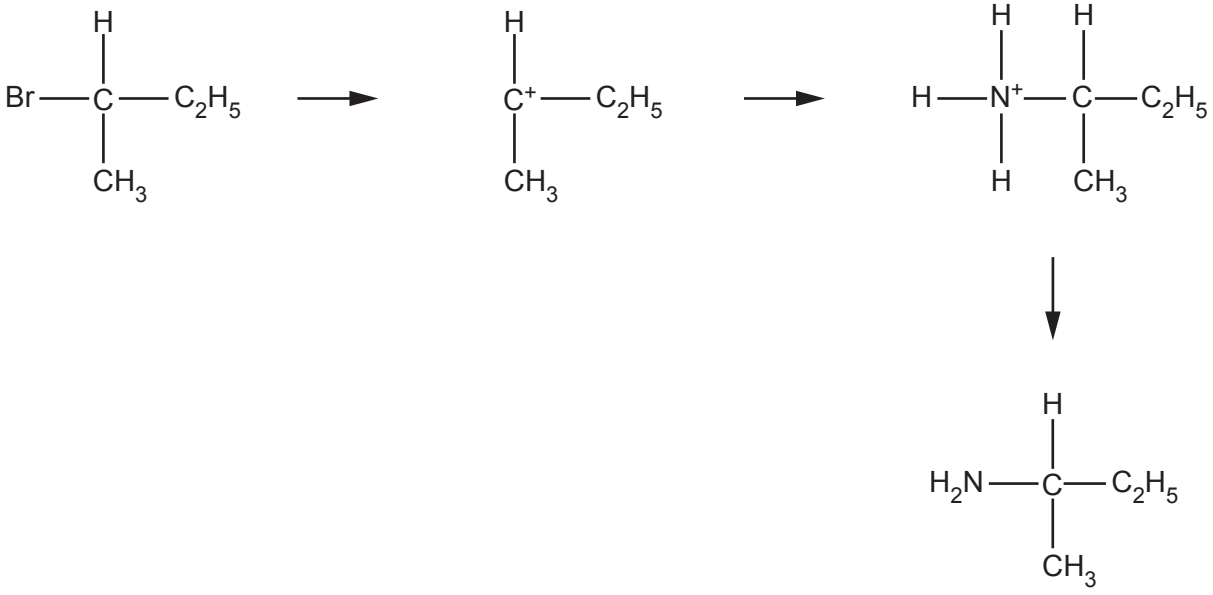


Fig. 4.2

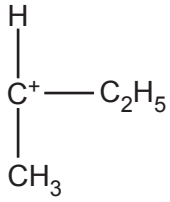
[3]

(ii) Identify the inorganic product that forms in the reaction in Fig. 4.2.

..... [1]

(f) **(i)** Complete Table 4.1 by drawing the structural formula of the intermediate that is formed when 2-bromo-2-methylpropane reacts in an S_N1 reaction.

Table 4.1

	2-bromobutane	2-bromo-2-methylpropane
structural formula of intermediate in S_N1 reaction		

[1]

(ii) Identify the halogenoalkane in Table 4.1 that has the greater tendency to react using the S_N1 mechanism. Explain your answer.

.....

.....

..... [2]

5 **(a)** **M** reacts to form **R** by the addition of one reagent, as shown in Fig. 5.1.

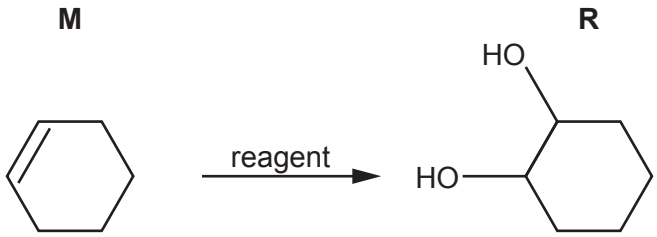


Fig. 5.1

Identify the reagent and conditions for this reaction.

..... [1]

(b) **R** is also made from **M** by two steps, as shown in Fig. 5.2.

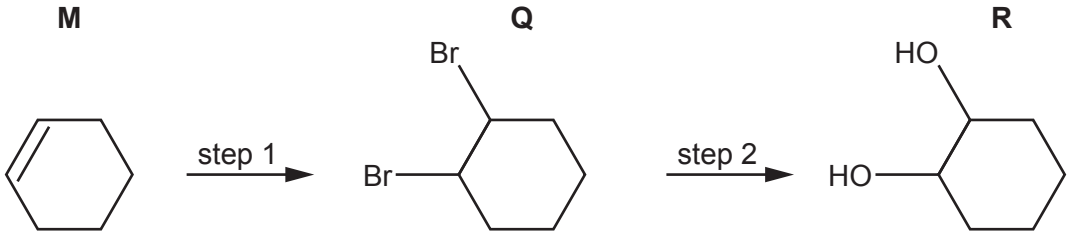


Fig. 5.2

(i) Identify the reagents and conditions for steps 1 and 2 in Fig. 5.2.

step 1 .....

step 2 .....

[2]

(d) **Y** is made from **Q** in a three-step reaction.

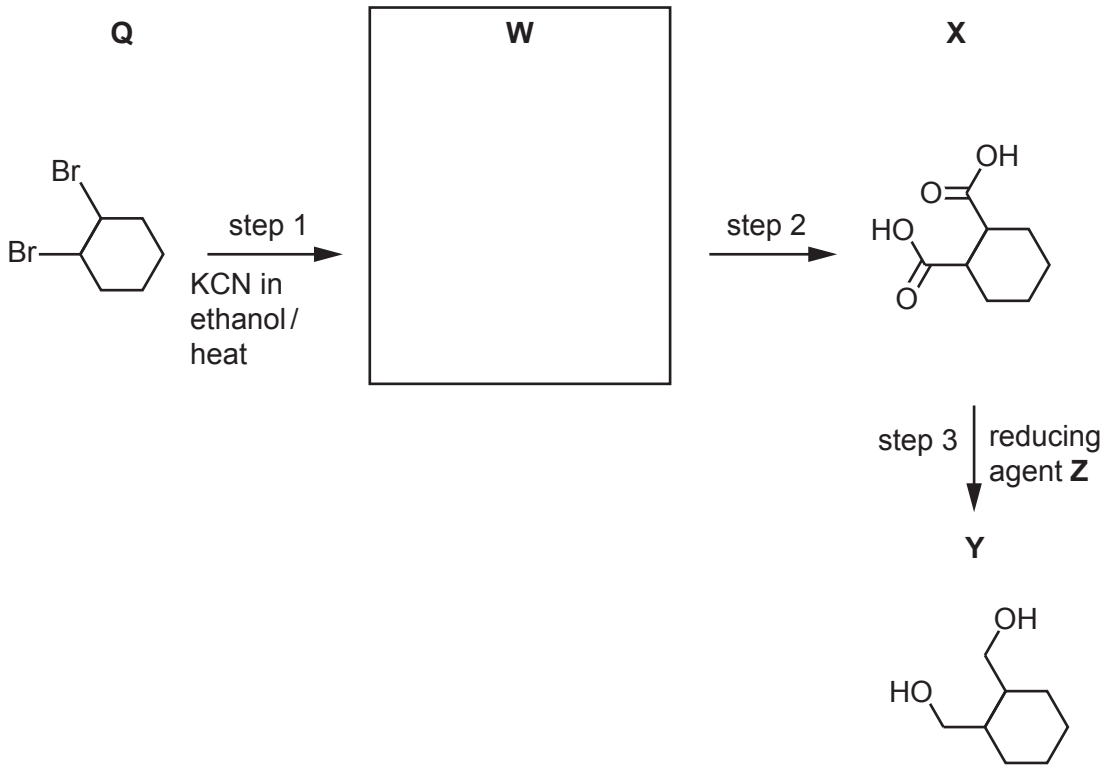


Fig. 5.4

(i) Draw the structure of **W** in the box in Fig. 5.4. [1]

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4 $CH_3(CH_2)_5CHBrCH_3$ exists as a pair of stereoisomers.

(a) Draw the three-dimensional structures of the **two** stereoisomers of $CH_3(CH_2)_5CHBrCH_3$. R can be used to represent $CH_3(CH_2)_5$.

[2]

(b) A sample of $CH_3(CH_2)_5CHBrCH_3$ reacts with NaOH to make $CH_3(CH_2)_5CH(OH)CH_3$ in an S_N1 mechanism.

Complete Fig. 4.1 to show the mechanism for the reaction of $CH_3(CH_2)_5CHBrCH_3$ and NaOH.

Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.



Fig. 4.1

[3]

(c) Separate samples of $CH_3(CH_2)_5CHBrCH_3$, $CH_3(CH_2)_5CH(OH)CH_3$ and $CH_3(CH_2)_5CHCH_2$ are tested with different reagents.

Complete Table 4.1. If no reaction occurs, write × in the relevant box.

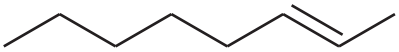
Table 4.1

reagent added	observation with $CH_3(CH_2)_5CHBrCH_3$	observation with $CH_3(CH_2)_5CH(OH)CH_3$	observation with $CH_3(CH_2)_5CHCH_2$
Br_2(l) in the dark			
PCl_5(s)			
$AgNO_3$(aq)			

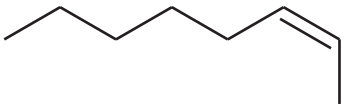
[3]

(d) $CH_3(CH_2)_5CHBrCH_3$ is heated with **D** to produce three different molecules, **E**, **F** and **G**.

E



F



G

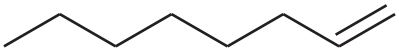


Fig. 4.2

(i) Name the type of reaction.

..... [1]

(ii) Identify **D** and the conditions used.

..... [1]

2 Chlorine, Cl_2, reacts with many elements and compounds to form chlorides.

Table 2.1 shows information about some chlorides of Period 3 elements.

Table 2.1

	Na	Mg	Si
formula of chloride			
structure of chloride	giant		
bonding of chloride			covalent
pH of solution formed on addition of chloride to water		6.2	

(a) Complete Table 2.1. [3]

(d) Cl_2 reacts readily with propene to form **K**, 1,2-dichloropropane.

K can be used to form **L**.

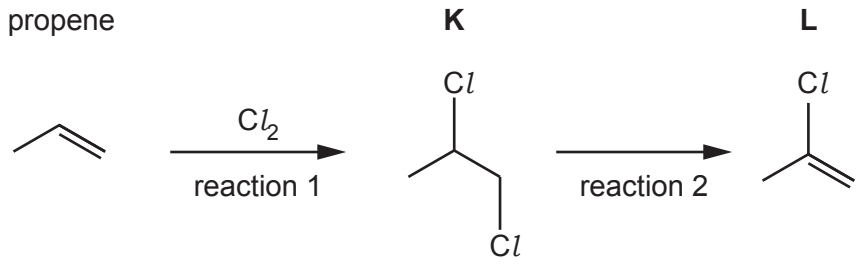


Fig. 2.1

(i) Complete Fig. 2.2 to show the mechanism for the reaction of Cl_2 with propene in reaction 1.

Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.

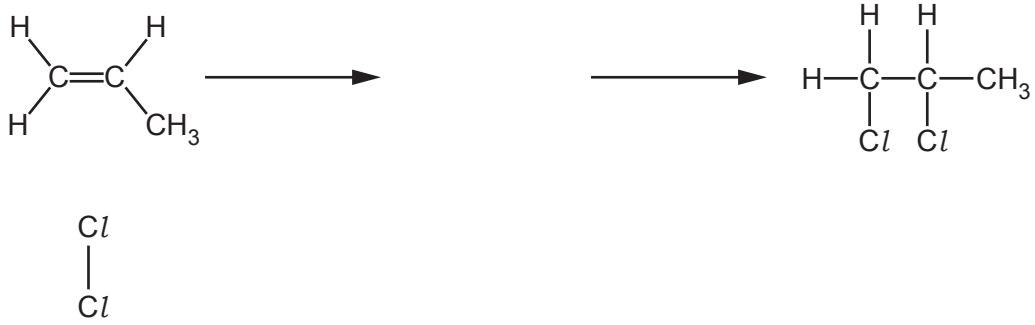


Fig. 2.2 [4]

(ii) Identify the reagent and conditions for reaction 2.

..... [1]

4 Compound **S** is used in food flavourings. A possible synthesis of **S** is shown in Fig. 4.1.

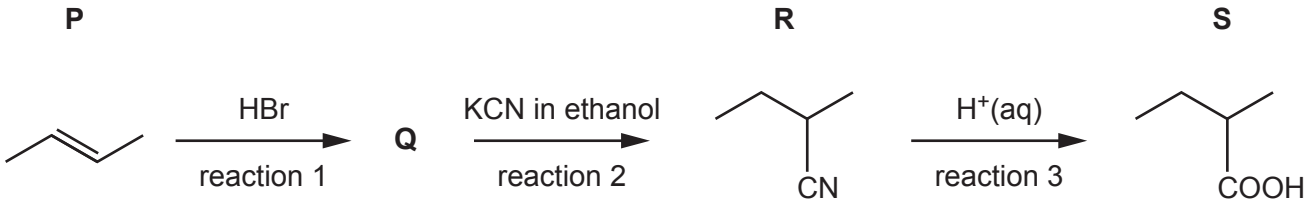


Fig. 4.1

(a) **P**, **Q**, **R** and **S** show stereoisomerism.

Complete Table 4.1 by identifying with a tick (✓) the type of stereoisomerism that each molecule shows.

The type of stereoisomerism shown by **Q** is given.

Table 4.1

	P	**Q**	**R**	**S**
geometrical isomerism				
optical isomerism		✓		

[2]

(b) **(i)** Give the structural formula of **Q**.

..... [1]

(ii) Name the mechanism in reaction 2.

..... [1]

3 Phosphoric(V) acid, H_3PO_4, is used in both inorganic and organic reactions.

(a) H_3PO_4 is made in a two-step process from phosphorus.

step 1 Phosphorus reacts with an excess of oxygen to form a white solid.

step 2 The white solid then reacts with water to form H_3PO_4.

(i) Write an equation for each step.

step 1 .....

step 2 .....

[2]

(c) Fig. 3.1 shows a reaction scheme that involves H_3PO_4 in several reactions.

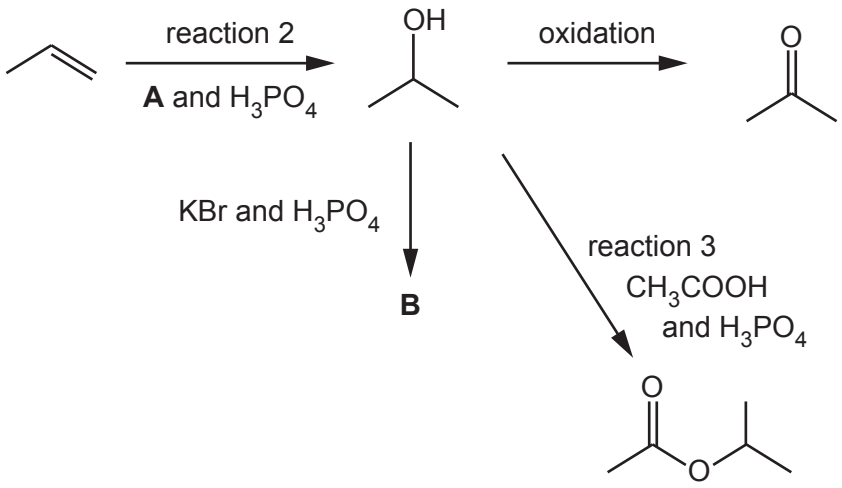


Fig. 3.1

(i) Identify **A**, which reacts with propene in the presence of H_3PO_4 in reaction 2.

..... [1]

(ii) Draw the structure of **B**.

[1]

6 Compound **W**, CH_2=CHCN, is used to make an addition polymer which is present in carbon fibres.

(a) Draw **one** repeat unit of the addition polymer of **W**.

[1]

(f) Propylamine can also be formed in a two-step synthesis from propan-1-ol, as shown in Fig. 6.4.

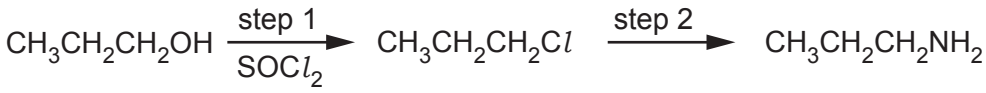


Fig. 6.4

(i) Name the type of reaction in step 1 in Fig. 6.4.

..... [1]

(ii) Identify the reagent and conditions for step 2 in Fig. 6.4.

.....

..... [2]

3 Fig. 3.1 describes a sequence of reactions that can be used to produce a food additive, compound **Y**, from CH_3CH_2Cl.

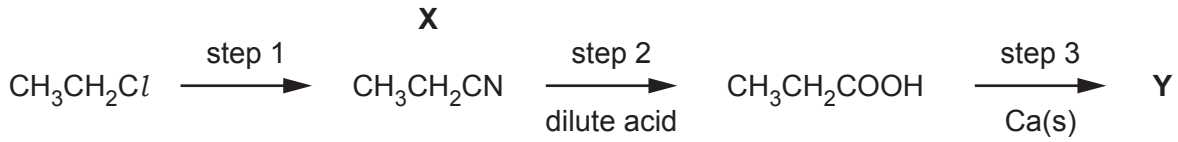


Fig. 3.1

(a) (i) State the reagent and conditions for step 1 in Fig. 3.1.

..... [1]

3 Chlorine is a very reactive element.

(a) Chlorine reacts with silicon to form silicon(IV) chloride.
Describe the appearance of silicon(IV) chloride at room temperature and pressure. State its structure and bonding.

appearance .....

structure and bonding .....
[2]

(d) Chlorine reacts with methane in a series of reactions to produce chloroalkanes.

(i) State the conditions required for chlorine to react with methane.

..... [1]

4 Fig. 4.1 shows some reactions of compound **D**, 2-bromobutane.

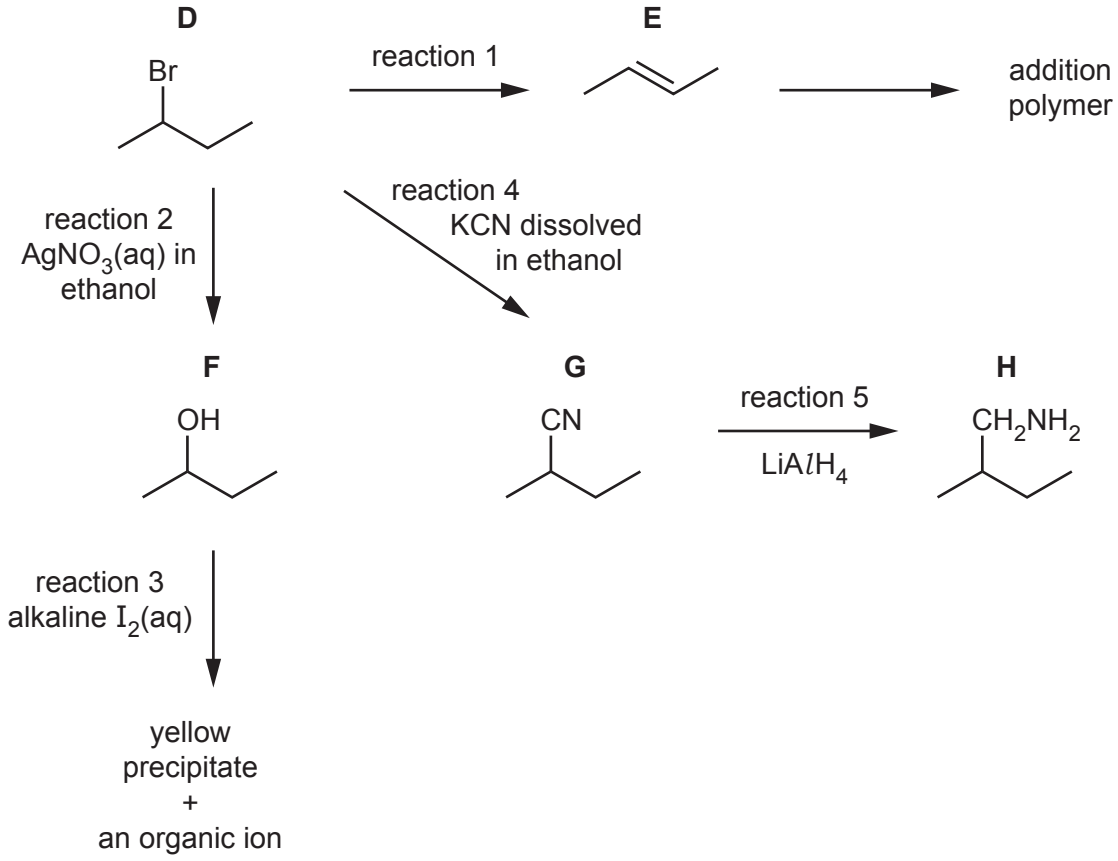


Fig. 4.1

(a) (i) State the reagent and conditions used to form **E** in reaction 1.

..... [1]

(b) (i) Predict what is observed in reaction 2.

..... [1]

(c) (i) State the type of reaction that occurs in reaction 4.

..... [1]